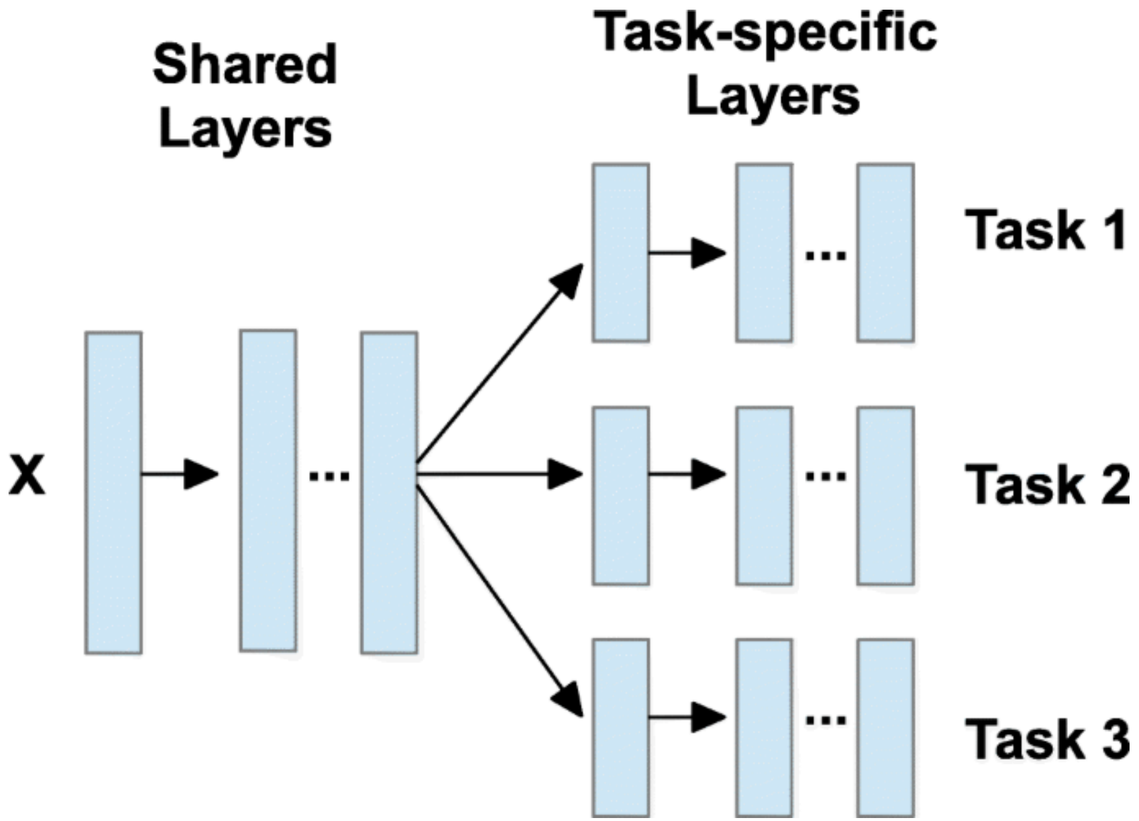
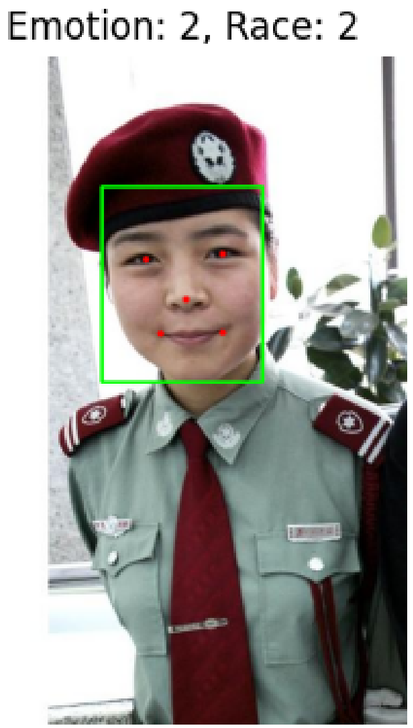


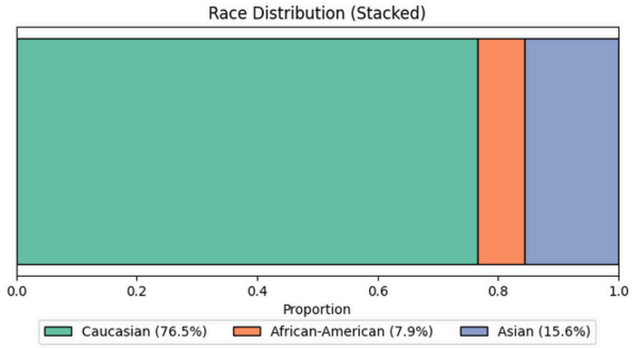
MULTITASK-LEARNING

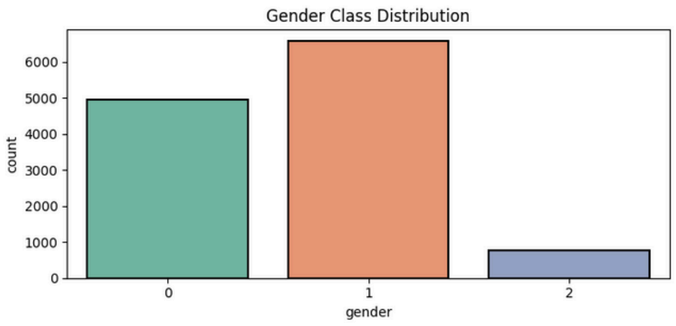
9 HEADS: EMOTION, AGE, RACE, GENDER, 5 FACIAL LANDMARKS

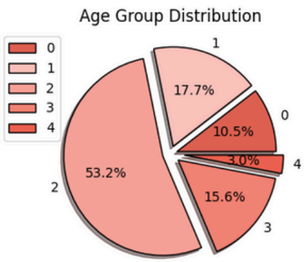
We wanted to try to achieve decent performance on a highly unbalanced dataset without using any pretrained backbones

AIM







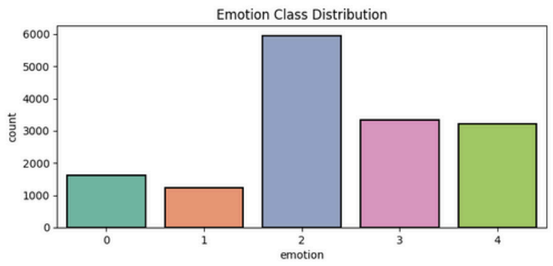


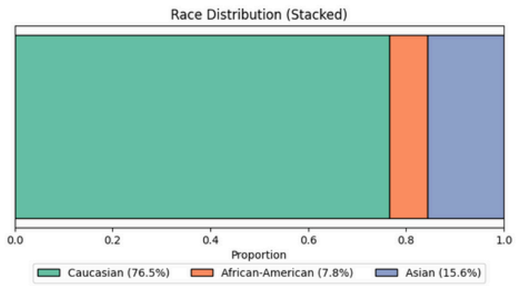
DATASET - REAL-WORLD AFFECTIVE FACES DATABASE (RAF-DB)

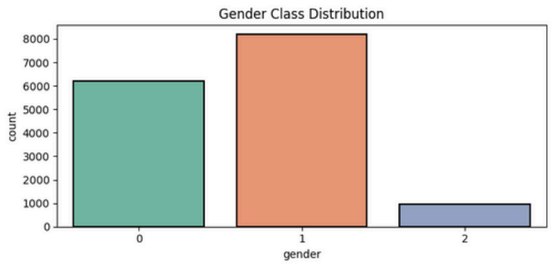
it was hard to balance stuff, especially emotions where majority class was 17X more than minority

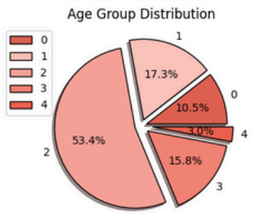
Same dataset after feature engineering.
More balanced but still not enough

Dataset contains 15339 samples,
7 emotion classes, 3 genders, 3races, and 5 age groups









DATASET HANDLING AND MODEL INITIALIZATION

TORCH.DATASET AND ALBUMENTATION FOR DATA HANDLING

Because we had to 1st load data, but also cut bbox, then adjust landmarks accordingly. Albumentations helped in that aspect. Then Normalize

Albumentation also helps apply other transforms automatically when data augmentation is applied to img

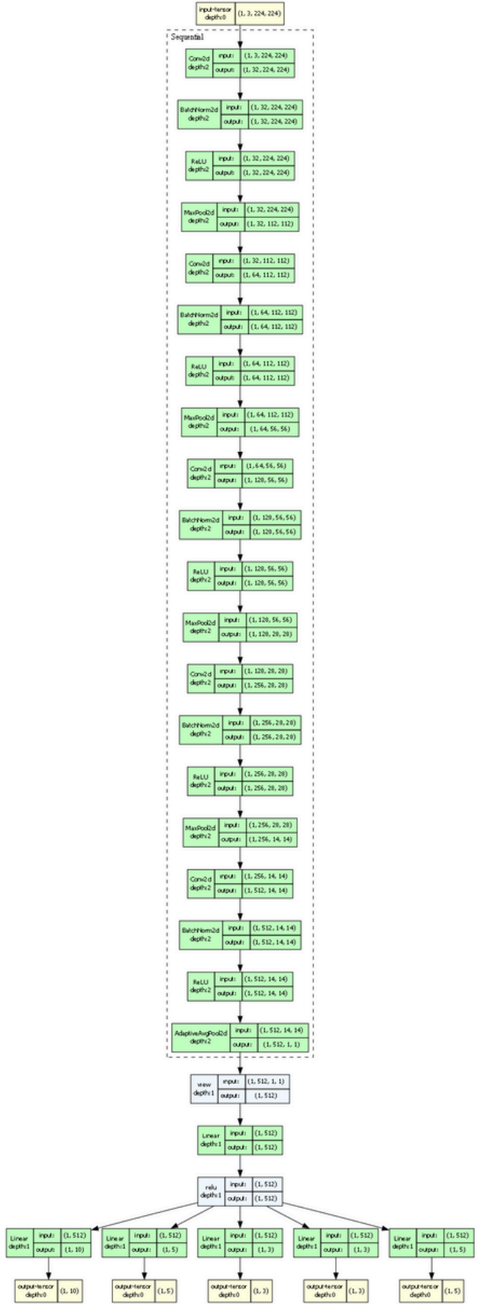
landmarks_manual	bbox	image_path
[[270.17, 330.696], [374.086, 302.67],	[213.92009, 199.133484, 484.330719,	./RAF/Images/original/train_00001.jp
[[532.365, 573.652], [753.071, 574.15	[407.148193, 265.544525, 912.43969	./RAF/Images/original/train_00002.jp
[[463.469, 554.446], [596.902, 515.02	[418.01532, 345.02063, 772.46106, 8	./RAF/Images/original/train_00003.jp
[[296.59, 316.573], [422.501, 318.766	[241.747635, 167.216034, 511.34378	./RAF/Images/original/train_00004.jp
[[161.18, 181.754], [221.991, 174.016	[114.065239, 113.609695, 249.96037	./RAF/Images/original/train_00005.jp

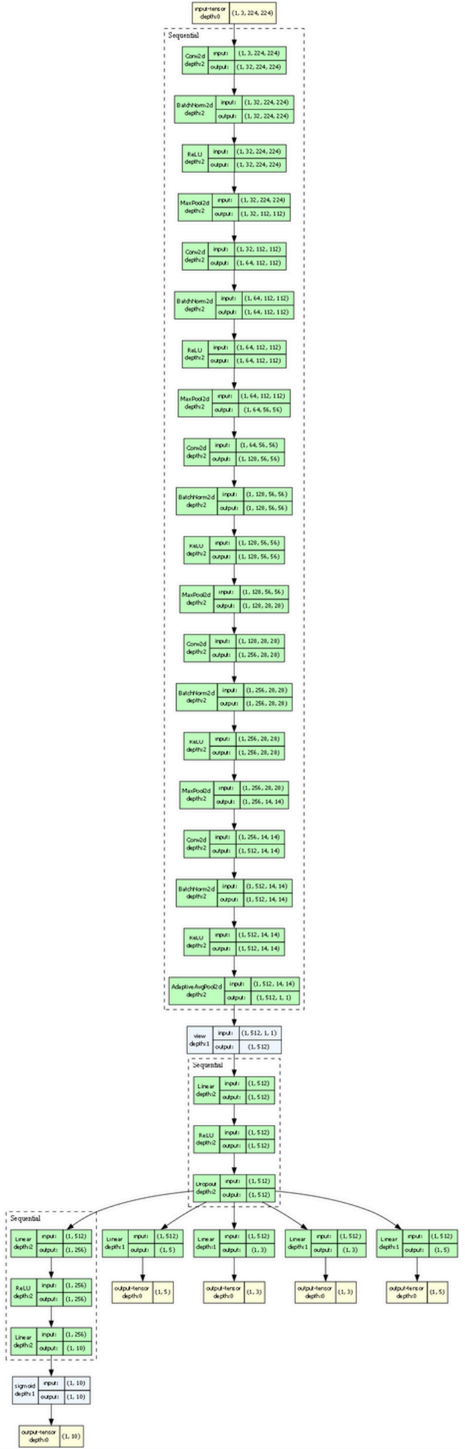
Model init

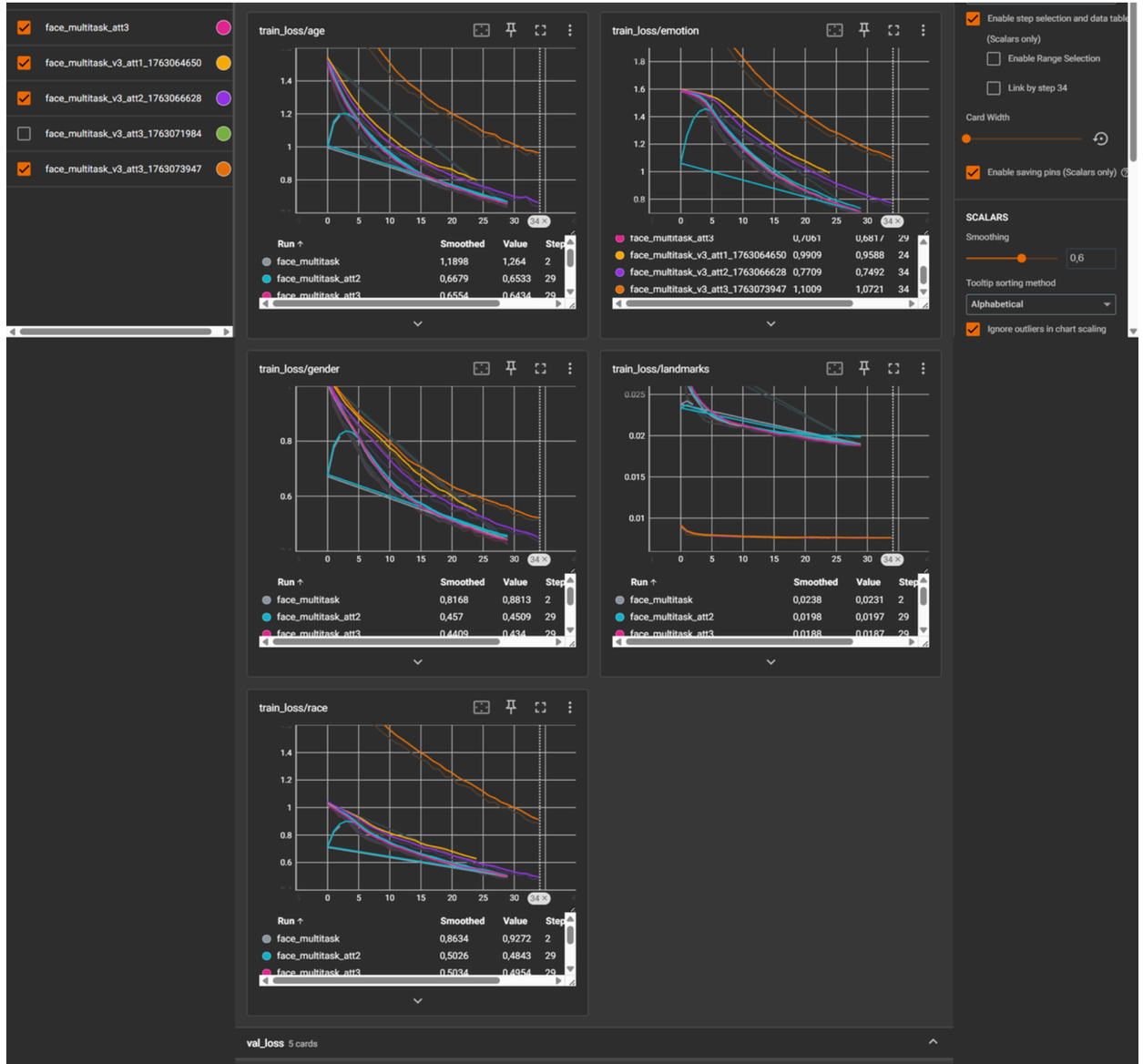
For categorical losses, we defined cross-entropy losses with weights, and MSE for regression head

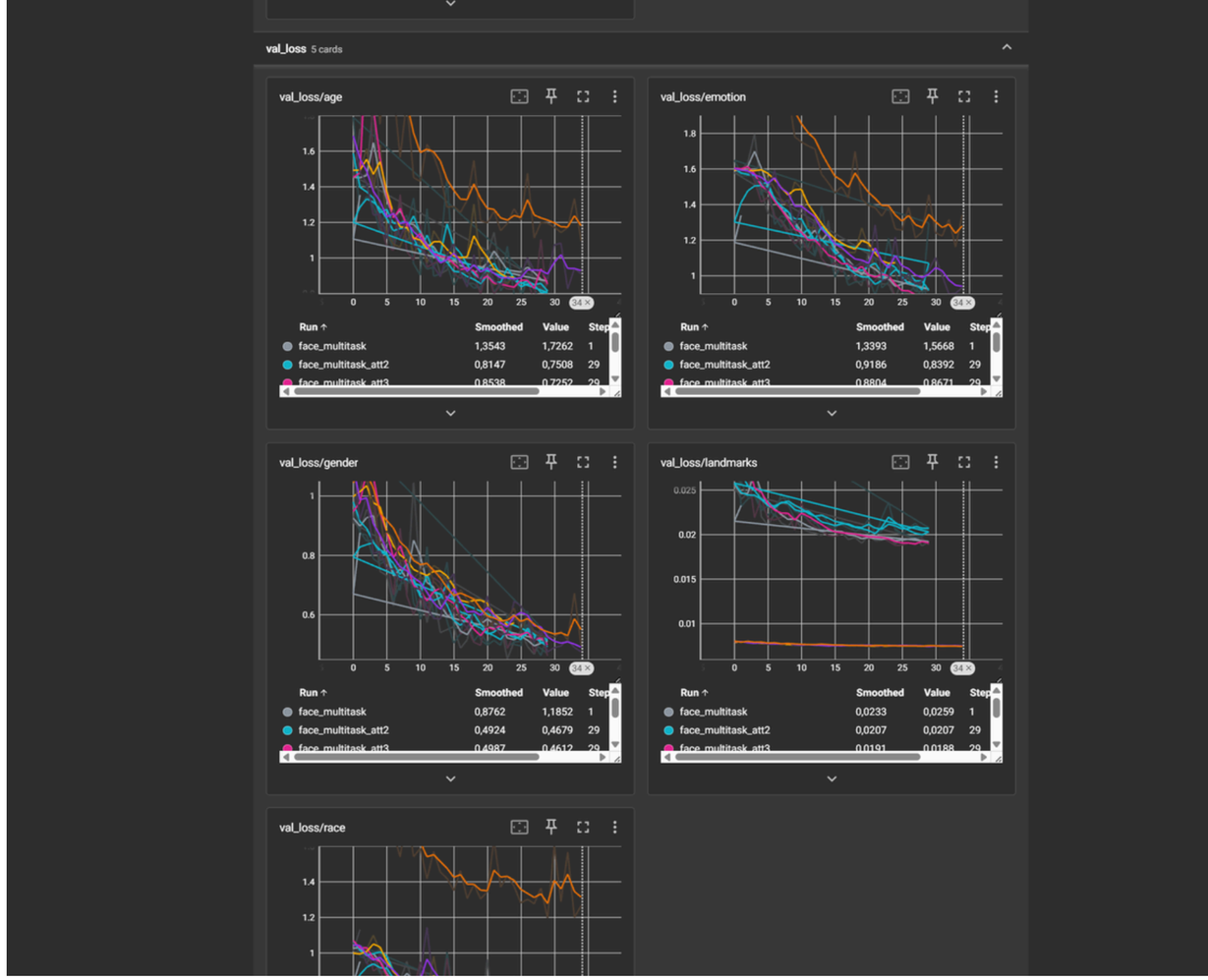
We tried AdamW along with Adam for optimizer, but Adam performed better

ARCHITECTURE









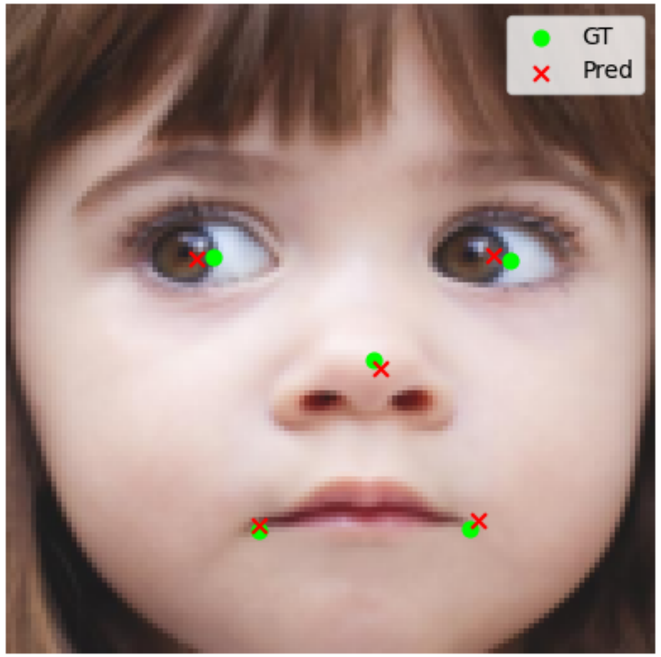
Train and validation loss history in our 2nd attempt

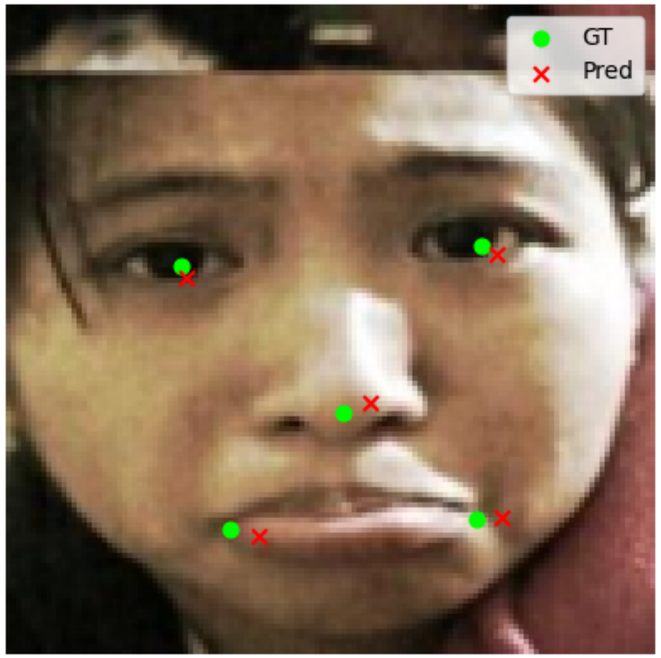
```
landmarks_mse: 0.0217
emotion_acc: 0.7018
emotion_f1: 0.7089
emotion_precision: 0.7389
emotion_recall: 0.7018
race_acc: 0.7480
race_f1: 0.7717
race_precision: 0.8316
race_recall: 0.7480
gender_acc: 0.7832
gender_f1: 0.7939
gender_precision: 0.8260
gender_recall: 0.7832
age_acc: 0.6369
age_f1: 0.6513
age_precision: 0.6949
age_recall: 0.6369
```

```
landmarks_mse: 0.0074
emotion_acc: 0.7005
emotion_f1: 0.7000
emotion_precision: 0.7265
emotion_recall: 0.7005
race_acc: 0.5962
race_f1: 0.6361
race_precision: 0.8137
race_recall: 0.5962
gender_acc: 0.7979
gender_f1: 0.8040
gender_precision: 0.8213
gender_recall: 0.7979
age_acc: 0.6786
age_f1: 0.6809
age_precision: 0.6849
age_recall: 0.6786
```

```
landmarks_mse: 0.0074
emotion_acc: 0.7008
emotion_f1: 0.7000
emotion_precision: 0.7304
emotion_recall: 0.7008
race_acc: 0.7174
race_f1: 0.7457
race_precision: 0.8224
race_recall: 0.7174
gender_acc: 0.7725
gender_f1: 0.7803
gender_precision: 0.8072
gender_recall: 0.7725
age_acc: 0.6444
age_f1: 0.6595
age_precision: 0.6983
age_recall: 0.6444
```

Correlation matrix will be show along with project demonstration on the screen



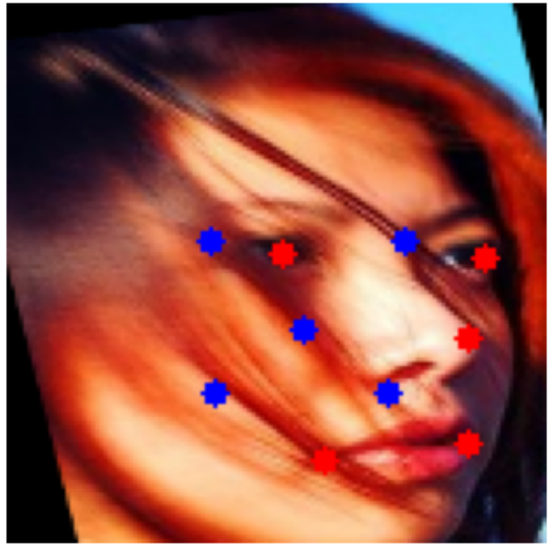


Attempt 1 vs Attempt 2

Overal categoricals saw a big leap, but landmarks started to struggle

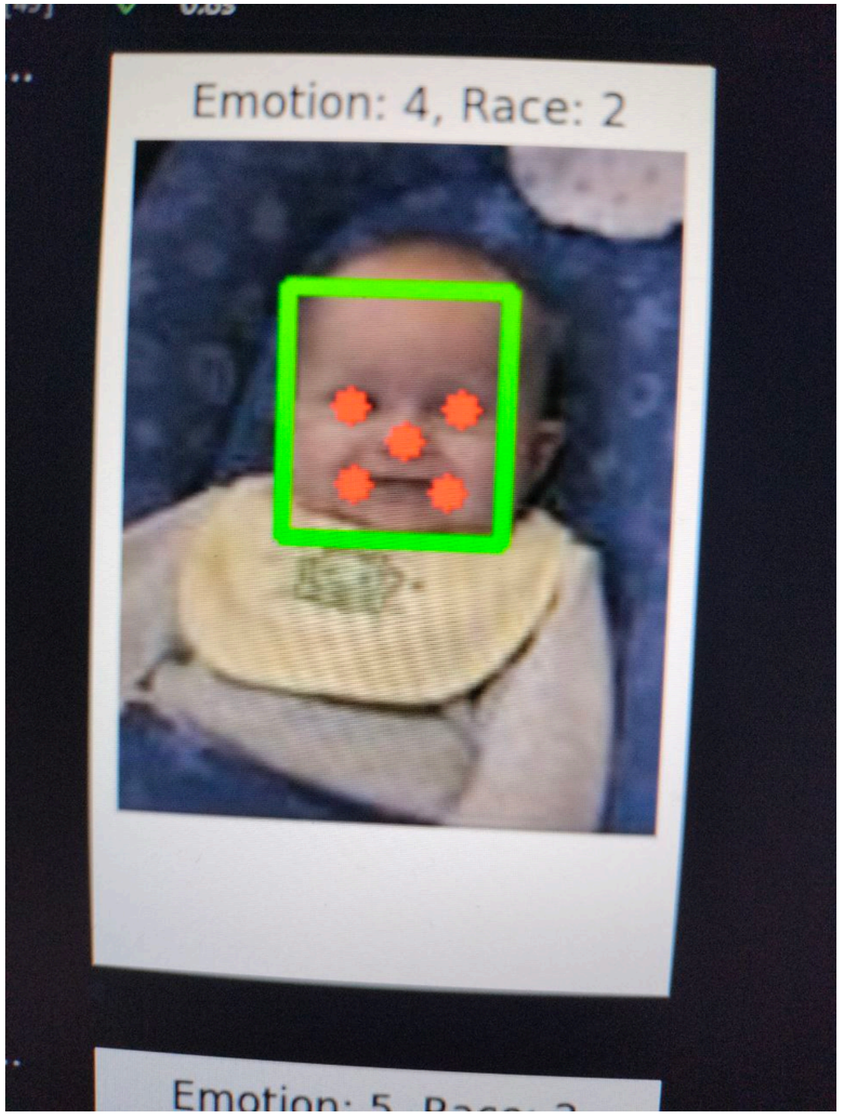
Both models kinda struggle when the person is not aligned, but it can be fixed with face crop tools

Emotion: 4, Race: 2, Gender: 1, Age: 2



Emotion: 4, Race: 1, Gender: 1, Age: 2

THANKS FOR ATTENTION

If you have any questions
feel free to ask

ASHIRBEK BEKZAT / MENIS MIRAS
BDA-2306